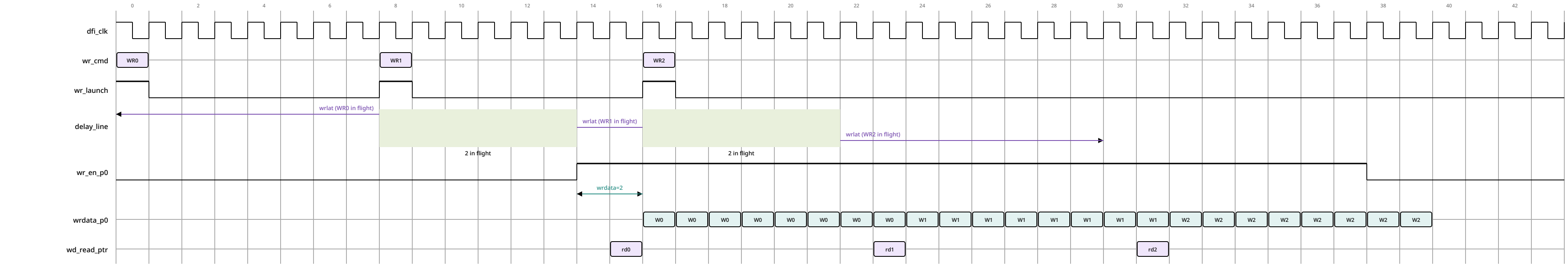


wr_launch needs a DELAY LINE (not one counter) - wrlat(14) > spacing(8)

WR0/1/2 @ 0/8/16; each launch injects a shift-reg entry; taps fire en@wrlat, wd_read_ptr@data. Pipes OVERLAP -> multiple in flight.



wrlat=14 > spacing=8 -> when WR1 launches (cyc8) WR0 is still in the WL pipe -> 2 entries coexist. one counter cannot; a shift-reg delay line of depth WL holds them all. real DDR5: WL~40 -> ~5 in flight.